

Nisrine KHNIFASS

Cybersecurity • Payment Security

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Final-year engineering student at ENSICAEN, specializing in cybersecurity and electronic payment security. Interested in application security and payment system protection. Currently studying abroad in Japan, seeking a six-month internship starting March 2027.

Cybersecurity and Software Projects

Magecart Detection Scanner (PCI DSS 4.0) - Personal project — 2026 - present
Python, Playwright, SHA-256 | Application security and e-commerce payment compliance

- **Developed prototype v0.1:** scans scripts loaded on payment pages, computes **SHA-256 hashes**, compares successive scans and checks allowed domains, aligned with PCI DSS 4.0 requirements (6.4.3 and 11.6.1).
- **Validation:** tested in a local environment with a simulated third-party script.

Project Lead - Interactive Escape Game - H3P Solution — 2026
Flutter | GitLab | real-time multi-team mobile app

- Led a **four-person team** to develop a real-time, multi-team escape game mobile app using **Flutter** and **GitLab**, delivering a **functional MVP approved by the client**.
- **Secure admin portal:** unique-identifier authentication, hashed passwords and session management.

Hands-on Network and Payment Security Labs - ENSICAEN — 2024 - 2026
Wireshark, ARP spoofing, EMV, APDU (ISO 7816)

- **Attack:** simulated a Man-in-the-Middle attack (ARP spoofing).
- **Cryptography:** cryptanalysis, Diffie-Hellman key exchange and password-strength testing.
- **Payments:** analyzed EMV transactions and communicated with smart cards via APDUs (ISO 7816).

Experience

Computer Vision Research Intern - SKKU — Apr. - Aug. 2026
Seoul, South Korea | Python, PyTorch, TensorFlow

- Developed a **five-module** Python video pipeline to detect household hazards to children from surveillance footage (approx. **48 frames/s**).
- **Evaluation: 88.7 % overall accuracy** and **100 % recall** for hazardous situations in the conducted tests.

Education

Shibaura Institute of Technology — Sep. 2026 - Feb. 2027
Tokyo, Japan | Computer Science Exchange Semester

ENSICAEN - Graduate School of Engineering, Caen — 2024 - 2027
Caen, France | Engineering degree in Computer Science, cybersecurity and payment systems

- Key coursework: cybersecurity, cryptography, software security, electronic payments, software engineering.

EM Normandie (dual degree) — 2025 - 2027
Caen, France | Master's Degree in Management

Lycée Carnot - Preparatory Classes (Honors) — 2022 - 2024
Dijon, France | Intensive preparatory program in mathematics and physics (PCSI - PC)

Skills

Applied in projects / labs | *Academic knowledge*

Cybersecurity: **AES, RSA, SHA-256** ; *ECC, PKI/X.509, TLS 1.3, IPsec/VPN, ProVerif*
Payments & banking: **EMV, ISO/IEC 7816** ; *PCI DSS, HSM, ISO 8583, ISO/IEC 14443*
Application & network security: **Web security, secure development, Wireshark, Nmap, OpenSSL**
Programming: **Python, Java, C, C++, JavaScript, Dart, Bash**
Frameworks & Data/IA : **React, Node.js, Flutter, PostgreSQL, PyTorch, TensorFlow**
Tools: **Git, GitLab, GitHub, Linux, Agile, SOLID**

Additional Experience and Languages

Sales Associate - C&A (2024 - 2025) : independent department management, inventory and customer service.

Rental Property Coordinator (2022 - 2024) : rental management, contracts and scheduling.

Languages: French: native | English: fluent - TOEIC 900 | Spanish: intermediate | German: beginner | Japanese: beginner

Additional information: Driving license (B), BAFA youth activity certificate | Interests: entrepreneurship, travel, sports, cooking